

AI-Human Collaboration for Optimal Point Dispersion in Unit Square
(n=16)

Abstract

We present a new world record solution for the Point Distance Ratio problem with $n=16$ points in a unit square. The objective is to minimize $r^2 = (\max_distance / \min_distance)^2$, where distances are computed between all pairs of 16 points constrained to lie within $[0,1]^2$. Our solution achieves $r^2 = 12.889229907694491$, improving upon the previous best-known result of $r^2 = 12.889246743$ established by FICO Xpress Global in 2025. This represents an improvement of 0.000016835. The optimization was performed using a hybrid evolutionary approach combining CMA-ES with IPOPOP restarts, Differential Evolution, and Basin Hopping, implemented through human-AI collaboration using Claude (Anthropic). The result was independently verified through four different computational methods, confirming the validity of our world record claim.

****Keywords**:** combinatorial optimization, point packing, evolutionary algorithms, CMA-ES, AI-assisted research

1. Introduction

1.1 Research Background

The Point Distance Ratio problem, also known as the dispersion problem, asks: given n points to place in a bounded region, how can we arrange them to minimize the ratio between the maximum and minimum pairwise distances? This problem has applications in facility location, sensor placement, experimental design, and telecommunications network planning.

For n points in a unit square $[0,1]^2$, the objective function is:

$$r^2 = \frac{\max_{i < j} d(p_i, p_j)}{\min_{i < j} d(p_i, p_j)}$$

where $d(p_i, p_j)$ denotes the Euclidean distance between points i and j .

The case $n=16$ is particularly interesting as it corresponds to a 4×4 grid structure, yet the optimal solution deviates significantly from a regular grid arrangement.

1.2 Research Objective

The primary objectives of this research are:

1. To surpass the existing world record of $r^2 = 12.889246743$ for $n=16$ points
2. To demonstrate the effectiveness of AI-human collaboration in solving hard optimization problems
3. To provide rigorously verified coordinates that can be independently reproduced

1.3 Research Hypothesis

We hypothesize that combining modern evolutionary algorithms (CMA-ES) with systematic multi-start strategies and AI-assisted algorithm design can discover solutions superior to those found by commercial mathematical programming solvers.

2. Related Work

2.1 Historical Background

The point dispersion problem has been studied extensively in the optimization literature. The problem is NP-hard in general, and finding global optima for even moderate values of n remains challenging.

2.2 Previous Approaches

Several approaches have been applied to this problem:

1. **Commercial Solvers**: FICO Xpress Global achieved $r^2 = 12.889246743$ using nonlinear global optimization (2025)
2. **LLM-based Discovery**: DeepMind's AlphaEvolve achieved $r^2 = 12.889266112$ using evolutionary code generation (2025)
3. **Classical Methods**: Previous best-known solutions achieved approximately $r^2 \approx 12.89$

2.3 AI in Optimization

Recent work has demonstrated the potential of AI systems in discovering novel algorithms and solutions. AlphaEvolve (Novikov et al., 2025) showed that LLM-powered evolutionary agents can match or exceed commercial solvers on geometric optimization problems.

3. Methodology

3.1 Problem Formulation

****Decision Variables****: 32 continuous variables representing $(x_1, y_1, x_2, y_2, \dots, x_{16}, y_{16})$

****Objective Function****:

...

minimize $r^2 = (\text{max_dist} / \text{min_dist})^2$
where $\text{max_dist} = \max\{d(p_i, p_j) : 1 \leq i < j \leq 16\}$
 $\text{min_dist} = \min\{d(p_i, p_j) : 1 \leq i < j \leq 16\}$
 $d(p_i, p_j) = \sqrt{[(x_i - x_j)^2 + (y_i - y_j)^2]}$
...

****Constraints****: $0 \leq x_i, y_i \leq 1$ for all $i = 1, \dots, 16$

3.2 Algorithm Design

We employed a multi-phase optimization strategy:

****Phase 1: Multi-Start CMA-ES****

- Algorithm: Covariance Matrix Adaptation Evolution Strategy (Hansen & Ostermeier, 2001)
- Initial configurations: 6 different starting points
 - Regular 4×4 grid
 - Hexagonal packing
 - Fibonacci spiral
 - Sobol quasi-random sequence
 - Halton sequence
 - D4-symmetric arrangement
- Population size: 48-256 (adaptive)
- Sigma (step size): 0.2 (decreasing)
- Termination: $\text{tolfun} = 10^{-12}$

****Phase 2: IPOP-CMA-ES (Increasing Population)****

- Strategy: Restart with $1.5 \times$ population size increase
- Maximum restarts: 50 per session
- 3 independent sessions

****Phase 3: Hybrid Refinement****

- Differential Evolution: $\text{maxiter}=20000, \text{popsize}=80, \text{mutation}=(0.4, 1.0)$
- Basin Hopping: $\text{niter}=30000, T=0.3, \text{stepsize}=0.08$
- L-BFGS-B: Final local refinement with $\text{ftol}=10^{-15}$

****Phase 4: Continuous Improvement****

- Infinite loop with periodic IPOP restarts and perturbation searches

3.3 AI Utilization

The AI assistant (Claude, Anthropic) contributed to:

1. Algorithm selection and parameter tuning recommendations
2. Code generation for optimization scripts
3. Debugging and error correction (e.g., Unicode encoding issues on Windows)
4. Verification script development
5. Documentation and report writing

The human researcher provided:

1. Problem specification and objectives
2. Strategic direction and phase planning
3. Execution monitoring and decision-making
4. Final verification and validation oversight

3.4 Implementation Environment

Component	Specification
CPU	Intel Core 12th Gen @ 2.1 GHz
RAM	16 GB
OS	Windows 11 Enterprise
Python	3.14.0
NumPy	2.4.1
SciPy	1.17.0
CMA-ES	4.4.1 (pycma)

3.5 Verification Methodology

Four independent verification methods were employed:

1. **scipy.spatial.distance.pdist**: Standard scientific computing library
2. **Pure Python (math module)**: No external dependencies
3. **NumPy-only**: Manual distance calculation
4. **Decimal (50-digit precision)**: High-precision arithmetic

4. Results

4.1 Optimization Result

Metric	Value
r^2	12.889229907694491
r (ratio)	3.590157365310674
Minimum distance	0.277397448421211
Maximum distance	0.995900492567799

| Number of distance pairs | 120 |

4.2 World Record Comparison

Source	Year	r^2	Value	Status
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Previous best-known	Pre-2025	~12.89	Superseded	
AlphaEvolve (DeepMind)	2025	12.889266112	Superseded	
FICO Xpress Global	2025	12.889246743	**Previous Record**	
This Work	**2026**	**12.889229907694491**	**New Record**	

Improvement over previous record: 0.000016835305509 (0.00013%)

4.3 Optimal Coordinates

Point	x	y
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1	0.751621065395568	0.508549292994639
2	0.997152523115845	0.379461541211772
3	0.871553887407651	0.132127097393604
4	0.323657580830225	0.355013309238508
5	0.591317355298941	0.282160103942124
6	0.818017674336209	0.877256389217622
7	0.055037343554039	0.702318753514680
8	0.986180066617098	0.656641896917198
9	0.583458673114692	0.729163785295059
10	0.624127418241828	0.006709846798874
11	0.323481394863156	0.632410701708421
12	0.243984724664492	0.905415359990766
13	0.117776964446627	0.169103341275221
14	0.346791957664821	0.012573863949459
15	0.506307262655133	0.995616415709360
16	0.016988892306614	0.427543100029373

4.4 Critical Point Pairs

****Minimum distance**** (0.277397448421211):

- Point 7: (0.055037343554039, 0.702318753514680)
- Point 11: (0.323481394863156, 0.632410701708421)

****Maximum distance**** (0.995900492567799):

- Point 8: (0.986180066617098, 0.656641896917198)
- Point 16: (0.016988892306614, 0.427543100029373)

4.5 Verification Results

Method	Computed r^2	Difference from Claim
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scipy.pdist	12.889229907694478	1.24×10^{-14}
Pure Python	12.889229907694478	1.24×10^{-14}
NumPy-only	12.889229907694478	1.24×10^{-14}
Decimal (50)	12.889229907694481	8.88×10^{-15}

All methods confirm the result within floating-point precision.

5. Discussion

5.1 Result Interpretation

The optimal configuration discovered deviates significantly from a regular 4×4 grid. The solution exhibits:

- Asymmetric point distribution
- 22 distance pairs within 1% of the minimum distance
- 8 distance pairs within 1% of the maximum distance

This suggests multiple near-minimum distances constrain the optimization, while the maximum distance is approached by multiple point pairs.

5.2 AI Contribution Analysis

The AI assistant contributed approximately:

- 80% of code generation
- 60% of algorithm design suggestions
- 90% of debugging assistance
- 95% of documentation

Human oversight was essential for:

- Strategic decision-making
- Execution monitoring
- Final verification approval
- Research direction

5.3 Limitations

1. **Global Optimality**: We cannot prove our solution is the global optimum. It represents the best-known solution.
2. **Numerical Precision**: Results are limited by IEEE 754 double-precision floating-point arithmetic.
3. **Reproducibility**: Stochastic algorithms may not reproduce identical results across runs.

5.4 Error Discovery and Correction

During development, we encountered:

1. ****Unicode Encoding Error****: Windows cp949 encoding failed on emoji characters. Fixed by replacing emojis with ASCII equivalents.
2. ****Function Definition Order****: NameError due to function called before definition. Fixed by reordering code.

6. Conclusion

6.1 Key Findings

We have established a new world record for the Point Distance Ratio problem with $n=16$ points:

- ****New record****: $r^2 = 12.889229907694491$
- ****Improvement****: 0.000016835 over FICO Xpress (2025)

6.2 Academic Contribution

1. First demonstration of AI-human collaboration breaking optimization world records in this problem class
2. Provision of fully verified, reproducible coordinates
3. Open-source verification toolkit for independent confirmation

6.3 Future Work

1. Extend methodology to larger n values ($n=17, 18, \dots$)
2. Apply similar techniques to 3D variants
3. Investigate theoretical bounds on achievable r^2 values
4. Develop hybrid AI-solver approaches combining LLMs with mathematical programming

References

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Appendix A: Verification Script

A standalone Python script (`verify\_record.py`) is provided for independent verification. It requires only NumPy and SciPy.

Appendix B: Computational Resources

- Total computation time: ~15 hours
- Primary algorithm: CMA-ES with IPOP restarts
- Hardware: Intel Core 12th Gen, 16GB RAM

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